

Study 5 Takeaways

1

Do not evaluate on training data

A decision tree can memorize the training set and report near-zero training MAE.

2

A single test split is only one sample

Changing `random_state` changes the measured test error.

3

Cross-validation is more informative

20-fold CV estimates $\text{MAE} = 0.46 \pm 0.01$ in \$100k units.

Conclusion: cross-validation turns model evaluation from a single fragile score into a distribution of evidence.